

MATLAB

Upload files · rajeshtavalam/EC/ x | Lab Manual - Classroom x | +

Ask Gemini

matlab.mathworks.com

School

HOME PLOTS APPS LIVE EDITOR INSERT VIEW

New Open Save Export Compare Go To Find Bookmark NAVIGATE

FILE

Normal

Text

Code Control Task Refactor

Code Issues Debugger

Generate Test Find Tests Run Section

Run Step Stop

ANALYZE TEST SECTION RUN

/ > MATLAB Drive

```
1 clc; clear; close all;
2 Ps = 1; % Signal Power (W)
3 Pn = [0.1 0.05 0.01]; % Noise Power (W)
4 Users = 1:3;
5 SNR = 10*log10(Ps./Pn); % SNR in dB
6 disp('User Noise Power(W) SNR(dB)')
7 disp([Users' Pn' SNR'])
8 figure
9 bar(Users,SNR)
10 xlabel('Users')
11 ylabel('SNR (dB)')
12 title('Signal-to-Noise Ratio of FDMA Users')
13 grid on
```

2.0000 0.0500 13.0103
3.0000 0.0100 20.0000

Signal-to-Noise Ratio of FDMA Users

Users	SNR (dB)
1	10.0
2	13.0
3	20.0

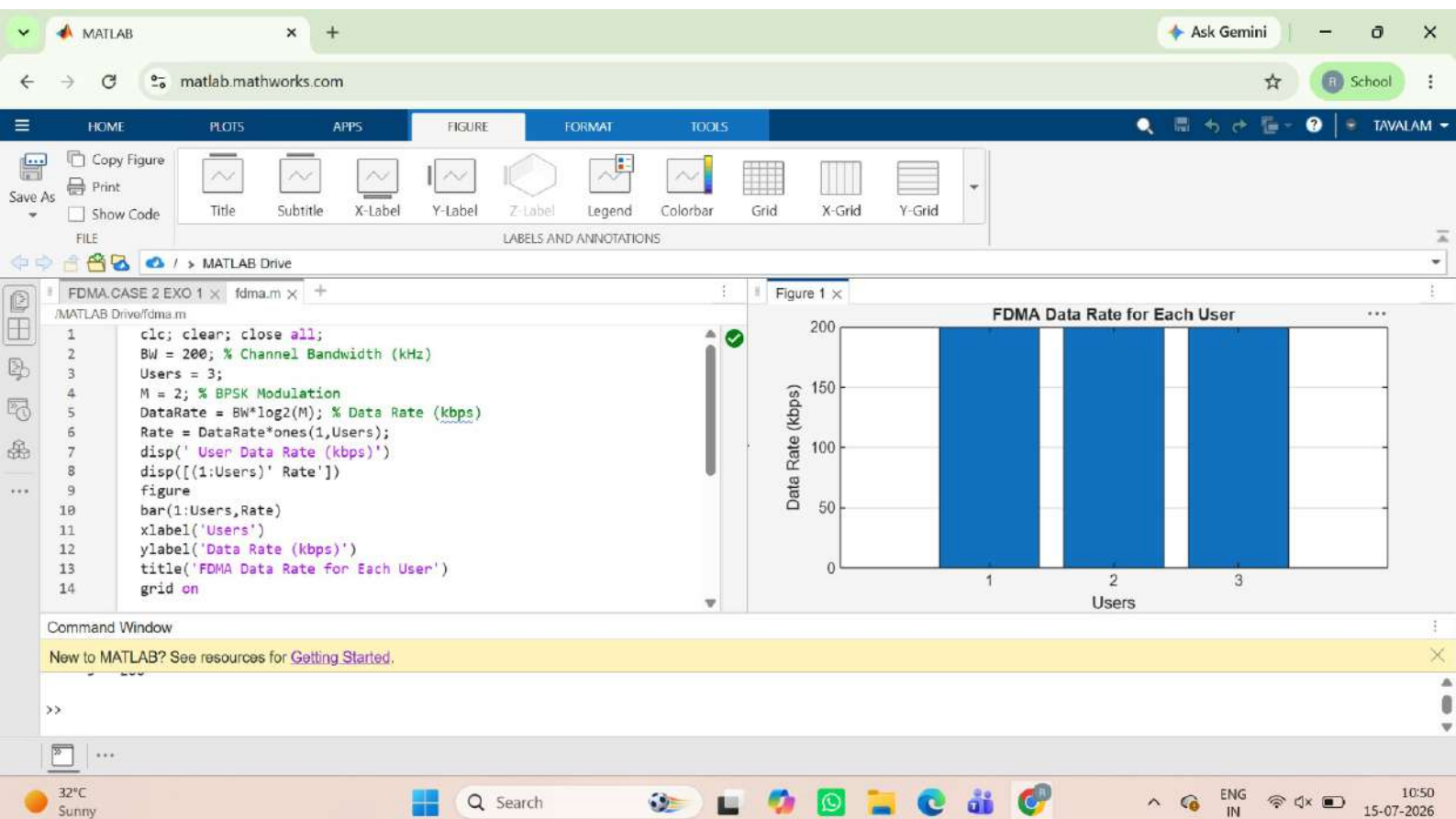
Editor: 100% UTF-8 LF Script Ln 13 Col 8

32°C Sunny

Search

ENG IN

10:56 15-07-2026



MATLAB

matlab.mathworks.com

HOME PLOTS APPS FIGURE FORMAT TOOLS

Save As Print Show Code FILE

Figure 1 x

```
1 clc; clear; close all;
2 fs=1000; t=0:1/fs:1;
3 TBW=600; Users=3;
4
5 CBW=TBW/Users;
6 fc=[100 300 500];
7 m=sin(2*pi*10*t);
8 s1=m.*cos(2*pi*fc(1)*t);
9 s2=m.*cos(2*pi*fc(2)*t);
10 s3=m.*cos(2*pi*fc(3)*t);
11 subplot(2,1,1)
12 plot(t,s1,'r',t,s2,'g',t,s3,'b'),grid on
13 title('FDMA Channel Allocation'),xlabel('Time (s)'),ylabel('Amplitude')
14 subplot(2,1,2)
15
```

Figure 1 x

FDMA Channel Allocation

Amplitude

Time (s)

Channel Bandwidth per User

Channel Bandwidth (kHz)

Users

Command Window

New to MATLAB? See resources for [Getting Started](#).

Feels hotter Now

Search

ENG IN

10:32 15-07-2026